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S-600 DRAINAGE AND DUCT WORKS

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S 600 DRAINAGE AND DUCT WORKS

1. SCOPE OF WORK

The work specified in this section covers design and supply of all labor, materials, transportation, equipment and services, installation or construction and testing for the following works as indicated in Part 4, Subsection 1.2 “Summary of Work “ and the Technical Specification S-001 “General Design Requirement of Structure, Civil and Building Works”.

- 1.1 Storm water drainage, sanitary water drainage, oily water drainage and waste water drainage shall include, but not limited to, ditches, piping, manholes, culverts etc.
- 1.2 Sanitary sewer system shall include septic tanks.
- 1.3 Cable duct, cable trench for electrical cables and pipes.
- 1.4 The excavation and the concrete work required for the above drainage and duct works shall be referred to Specifications S-300 “Earth Works” and S-340 “Concrete Works” respectively.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.
- 2.2 The Codes and Standards of the following organizations extracted from the Part 4, Subsection 2.5 “Codes and Standards” shall be specifically applicable to the design of the drainage and duct works included in the respective Technical Specification.
 - AASHTO..... American Association of State Highway and Transportation Officials.
 - ASTM.....American Society for Testing and Materials.
 - JIS..... Japanese Industrial Standards.

3. (not used)

4. STORM WATER DRAINAGE WORKS

Storm water drainage works shall comprise of culverts, ditches, pipes and manholes.

4.1 Concrete Ditch

Concrete ditches shall be cast in site or precast concrete at the location as shown on the Bid Drawings or as directed by the Owner. Construction of the concrete ditches shall be in conformance with lines, grades and dimensions as shown on the Bidder's construction drawings.

4.2 Reinforced Concrete Pipe

4.2.1 Pipe and Fittings: Reinforced concrete, confirming to the applicable requirements of ASTM C76, Class IV or approved equivalent.

4.2.2 Size: Subject to the Bidder's design.

4.2.3 Markings: Each length of pipe shall be clearly marked with name of manufacturer, location of plant where made, date of manufacture and pipe class.

4.2.4 Joints: The Bidder shall provide all materials required to complete the work and shall install rubber gasket or joint material between the joints of concrete pipe sections in strict accordance with the manufacturer's instructions.

4.2.5 Concrete pipe shall not be backfilled until inspected and approved by the Owner.

4.3 Culvert

Culverts shall be of reinforced concrete and if located under roads they shall be designed to withstand an HS20-44 loading. Expansion joint shall be provided for the culverts having more than 15 m length and the water stops as specified in Technical Specification S-340 "Concrete Works" shall be used.

4.4 **Manholes**

The manholes shall be made of precast concrete, complete with cast iron frames and covers, steps etc. specified herein. The manhole shall be made within maximum distance 50 meters.

4.4.1 **Precast Concrete Manholes**

- 1) Type: Precast reinforced concrete ring type with precast reinforced concrete base and rubber “O-ring” or flat type rubber compression joints, conforming to applicable requirements of ASTM C478, “Specifications for Precast Reinforced Concrete Manhole Risers” or approved equivalent.
- 2) Loading: Conforming to HS20-44 truck loading.
- 3) Precast Elements:
 - a. Base : Closed end pipe type with bell end.
 - b. Ring sections for walls: Multiples of 200 mm in each height of ring with tongue and groove joints.
 - c. Top section shall be of the eccentric cone type, or shall be the flat slab type with not less than 200 mm thick.
 - d. The rings and top cone shall have precast openings for field installation with galvanized iron steps and for all required drain pipes entering manholes.
- 4) Joints: Rubber “O-ring” or flat type rubber compression type, with manufacturer’s standard rubber ring. Mortar joints may be used only per the Owner’s approval.

4.4.2 **Frames, Covers and Steps**

- 1) Cast iron as indicated on the approved drawings.
- 2) Coating: Coat uniformly with coal tar pitch varnish.
- 3) Covers shall be with Reinforced Concrete, checker plate and steel grating.

- 4) Type: As indicated on the approved drawings.

4.4.3 **Installation of Manholes**

- 1) Subgrade shall be level and free of projecting stones, rocks etc.
- 2) Place a layer of sand, not less than 100 mm thick, over the subgrade before installing the precast base. Exercise care in installing the base level with full bearing throughout on the sand cushion to insure that the completed catch manhole shall be plumb.
- 3) Installing sections with rubber rings shall be executed in strict accordance with manufacturer's instructions.

5. **SANITARY WATER DRAINAGE**

- 1) All drainage from the kitchen and toilet areas shall be routed to the septic tank. Adequate air vent in the building plumbing system shall be provided.
- 2) The sanitary water drainage shall consist of manholes, cleanouts, piping, grease trap and septic tank as well as absorption field where required.
- 3) All sewers shall be designed to flow half full at normal flow with a minimum of 0.75 and maximum of 3.0 meters per second velocity. ALL sewers shall have a minimum cover of 0.75 meters.
- 4) Cleanouts shall be provided at the junction of the building sewer and at each change in direction of the sewer. If the sewer length from the building to the septic tank is more than 15 m, a cleanout shall be provided at minimum 15 m intervals. If the sewer line crosses any potable water line, 5 meters on both sides of the water line shall be of cast iron. If the sewer line is within 3 meter distance of the potable water line, the sewer shall be of cast iron pipe with leaded joints.

Provide grease trap in waste line from kitchen.

5.1 **Septic Tank**

- 1) Septic tanks shall have a sufficient capacity for accommodation of the possible number of people. Tanks shall be made of reinforced concrete

with manholes and steps and shall be of a easy operation, inspection and cleaning. The structures shall have sufficient strength against soil pressure, water pressure, load, etc. The inside surface of the tank and baffle walls shall be coated with bituminous coating.

- 2) The piping from the building to the septic tank shall be of cast iron or vitreous clay pipe. The minimum pipe size shall be 150 mm inside diameter.

5.1.1 **Mechanical Sewage System**

- 1) Mechanical sewage system shall consist of blower unit, pretreatment tank and main tank, and shall contain the following equipment.
 - a. Blower motor
 - b. Air diffuser
 - c. Comminutor
 - d. Flow control float
 - e. Surface skimmer
 - f. Sludge recirculation
 - g. Raw sewage pump
 - h. Effluent pump
 - I. Defoaming pump
 - j. Chlorinator
 - k. Control panel, piping and required pertinent.
- 2) The blower house shall be made of a reinforced concrete frame and roof slab, with concrete block masonry wall. The minimum roof height shall be 2.5 m.
- 3) Allowable BOD concentration for discharge water shall be 60 ppm.

5.1.2 **Conventional Septic Tank**

The septic tank shall consist of an inlet compartment and an outlet compartment with minimum depth of 150cm. Tank capacity of less than 3 m3 may have only one compartment. The tank shall be designed for sludge accumulation of minimum 10 years.

The discharge pipe shall be routed to the absorption field.

5.2 Absorption Field

- 5.2.1 Prior to the design of the tile drainage absorption field, the Bidder shall perform percolation tests to determine the absorption rate. Based on the field percolation test results the size of the absorption field shall be determined. The location of the absorption field shall be proposed by the Bidder for approval of the Owner.
- 5.2.2 The minimum depth of absorption field trenches shall be 450 mm. Minimum widths shall not exceed 750 mm. The individual laterals shall be preferably 20 m long but not longer than 30 m and the slope of the pipe in the trench shall not exceed 150 mm per 30 m. The trench filter material should be clean coarse sand. The sand preferably should have effective size between 0.4 and 0.6 mm and a uniformity coefficient not greater than 4. The pipes shall be of perforated vitreous clay pipe of 150 mm in diameter.

6. OTHER DRAINAGE SYSTEM

- 6.1 Oily Water Separator System
- 6.1.1 Oily and contaminated waste water from floor washing and other processes shall be discharged into the drainage system via an oil/water separator which ensures that the oil content in the waste water discharge is within statutory control limits.
- 6.1.2 The Bidder shall design and construct suitable numbers of appropriate sized oil/water separator sumps in reinforced concrete (Grade 25, cement type 2), at appropriate locations. Heavy duty cast iron covers including ventilation pipe with suitable hood shall be provided.
- 6.1.3 The invert levels of incoming/outgoing pipes shall be designed to complement the overall drainage system with respect to the areas being drained.
- 6.1.4 The basic concept of the oil/water separator system to be incorporated shall be submitted to the Owner in the Bid Proposal.
- 6.1.5 A device for measuring oil retained depth shall be included.

6.2 Chemical Waste Water Drainage System

6.2.1 Chemical waste water drainage system shall be referred to the Technical Specification M-404 Waste Water Treatment System.

6.2.2 The works in this Specification shall cover drainage ditch from neutralizing pit to the waste water treatment plant.

7. CABLE DUCT AND CABLE TRENCH

7.1 All cable ducts and cable trenches shall be reinforced concrete structure.

7.2 The ducts and trenches shall be poured in place reinforced concrete construction. The typical ducts and trenches are shown on the Bid Drawings for reference. All trenches shall be covered with removable trench covers. All trenches shall be drained. Pumps for draining shall be automatic permanent type.

7.3 The covers shall be precast reinforced concrete and shall have a smooth and even upper surface, be square in shape, free from cracks and from damage to edges.

7.4 At all changes in direction, thrust blocks having a minimum volume of 0.20 m³ per pipe or larger as required by design shall be provided and shall be cast in situ, centered on the bisector of the bend angle.

Sectional steel for support of covers shall be embedded in concrete. It shall be thoroughly set in line and level and duly fixed to the form and/or maintained as to avoid any displacement during pouring. Tolerance of setting (distance between vertical flanges) shall be + 0.5 cm (5mm) and - 0.00.

7.5 Draw pit shall be provided in duct not exceeding 8 m in length.

7.6 The ends of non-continuous cable or pipe ducts shall be temporarily sealed to prevent the ingress of backfill material.

7.7 In reinforced concrete cable ducts, the reinforcement shall be installed as not to form a closed loop around the ducts.

Individual reinforcement bars or welded mesh, which would form part of closed loop, shall be electrically insulated from each other in order to break the loop.

- 7.8 Unless otherwise required, all steel embedded in concrete shall be galvanized.

8. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

- 8.1 Quality control shall conform to the requirements of Part 4, Subsection 2.7.

- 8.2 The quality control procedures and documentation.

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